



**GyaanVault**

SPPU M.Sc. Data Science — Free Study Material

# Database Technologies

**DS-551-MJ**

Interview Preparation — Topic-wise Q&A

Semester II · Core

Curated interview questions and answers for Database Technologies, organised by topic from the SPPU M.Sc. Data Science syllabus. Ideal for viva, placement and internship preparation.

*Tip: read the answer, then close the PDF and try to explain each concept aloud in your own words.*

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## SQL & transactions

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### **Q. What are ACID properties?**

Atomicity, Consistency, Isolation, Durability — guarantees that database transactions are processed reliably.

### **Q. Difference between DDL, DML and DCL?**

DDL defines schema (CREATE, ALTER, DROP); DML manipulates data (INSERT, UPDATE, DELETE, SELECT); DCL controls access (GRANT, REVOKE, COMMIT, ROLLBACK).

### **Q. What is normalization?**

Organising data to reduce redundancy and anomalies, through normal forms (1NF–BCNF).

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## NoSQL

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### **Q. SQL vs NoSQL — when to use which?**

SQL suits structured data with fixed schema and complex joins; NoSQL suits large-scale, flexible-schema, high-velocity data (documents, graphs, key-value).

### **Q. What are aggregate data models?**

In NoSQL, data is grouped into aggregates (self-contained units like a document) that are read/written as a whole, aiding distribution.

### **Q. What is Map-Reduce?**

A programming model that processes large datasets in parallel: Map emits key-value pairs, Reduce aggregates values by key.

### **Q. What is polyglot persistence?**

Using different database technologies for different needs within one application (e.g. MongoDB for documents, Neo4j for relationships).

